Unit 02: Installation Guide

CONTENTS

Objectives

Introduction:

2.1 Booting sequence:

2.2 Download Linux

2.3 Installation of Linux

2.4 Moving around the Desktop

2.5 Components of Desktop

Summary:

Keywords:

Self Assessment

Answers for Self Assessment

Review Questions:

Further Readings

Objectives

After studying this unit, you will be able to

- Understand the booting process
- Understand the installation process of Linux
- Understand the partitioning of hard drives
- Understand the file system types
- Understand how to log in the system

Introduction: An operating system (OS) is a system software that manages computer hardware, software resources, and provides common services for computer programs. Booting is a bootstrapping process that starts operating systems when the user turns on a computer system. A boot sequence is the set of operations the computer performs when it is switched on that load an operating system.

2.1 Booting sequence:

The booting sequence follows several steps. These are:

- Turn on the system
- CPU jump to address of BIOS
- BIOS runs POST (Power-On Self-Test)
- Find bootable devices
- Loads and execute boot sector form MBR
- Load OS

The first and foremost task is to turn on the system. So that the other processes can start. Rest of the process includes jumping of CPU to the address of CPU, BIOS runs POST, finding of bootable devices, loads and execute boot sector from MBR and loading of operating system.

BIOS (Basic Input/ Output System)

BIOS refer to the software code run by a computer when first powered on. It identifies your computer's hardware, configures it, tests it, and connects it to the operating system for further instruction. This is called the boot process. The primary function of BIOS is code program embedded on a chip that recognizes and controls various devices that make up the computer.

MBR(Master Boot Record)

OS is booted from a hard disk, where the Master Boot Record (MBR) contains the primary boot loader.The MBR is a 512-byte sector, located in the first sector on the disk (sector 1 of cylinder 0, head 0).After the MBR is loaded into RAM, the BIOS yields control to it.

Boot loader

Boot loader could be more adeptly called the kernel loader. The task at this stage is to load the Linux kernel.GRUB and LILO are the most popular Linux boot loader.Examples of boot loaders are:

Example: GRUB, LILO, GRUB2WIN, BOOTCAMP, BOOTKEY, NTLDR and Syslinux

GRUB:

GRUB stands for GRand Unified Bootloader.It is an operating system independent boot loader. It is a multiboot software packet from GNU.It has a flexible command line interface. It has file system access. It supports multiple executable formats. It supports diskless system.

LILO: LInux Loader

This boot loader does not depend on a specific file system.It can boot from hard-disk and floppy

Task of kernel

The kernel helps in process management, memory management. The device management is also one of the tasks of kernel.The system calls are also handled by kernel.

Init process

The first thing the kernel does is to execute init program. Init is the root/parent of all processes executing on Linux. The first processes that init starts is a script /etc/rc.d/rc.sysinit. Based on the appropriate run-level, scripts are executed to start various processes to run the system and make it functional. The init process is identified by process id "1".Init is responsible for starting system processes as defined in the /etc/inittab file.Upon shutdown, init controls the sequence and processes for shutdown.

Runlevels

A run-level is a software configuration of the system which allows only a selected group of processes to exist. Init can be in one of seven run-levels: 0-6.

Runlevel	ScriptsDirectory (RedHat/Fedora Core)	State
0	/etc/rc.d/rc0.d/	shutdown/halt system
1	/etc/rc.d/rc1.d/	Single user mode
2	/etc/rc.d/rc2.d/	Multiuser with no network services exported
3	/etc/rc.d/rc3.d/	Default text/console only start. Full multiuser
4	/etc/rc.d/rc4.d/	Reserved for local use. Also X-windows (Slackware/BSD)
5	/etc/rc.d/rc5.d/	XDM X-windows GUI mode (Redhat/System V)
6	/etc/rc.d/rc6.d/	Reboot
s or S		Single user/Maintenance mode (Slackware)
M		Multiuser mode (Slackware)

2.2 Download Linux

- To install Red Hat, you will need to download the ISO images (CD Images) of the installation CD-ROMs from http://fedora.redhat.com
- Download the i386 images for 32 Intel Processors, PPC images for Apple Macintosh and x86_64 for 64-bit AMD Processors.

Here the installation of Linux distribution is shown under VMWare. It can be installed as an individual operating system as well. Generally, when we buy a new system, the seller gives us the Windows operating system by default in that.

2.3 Installation of Linux

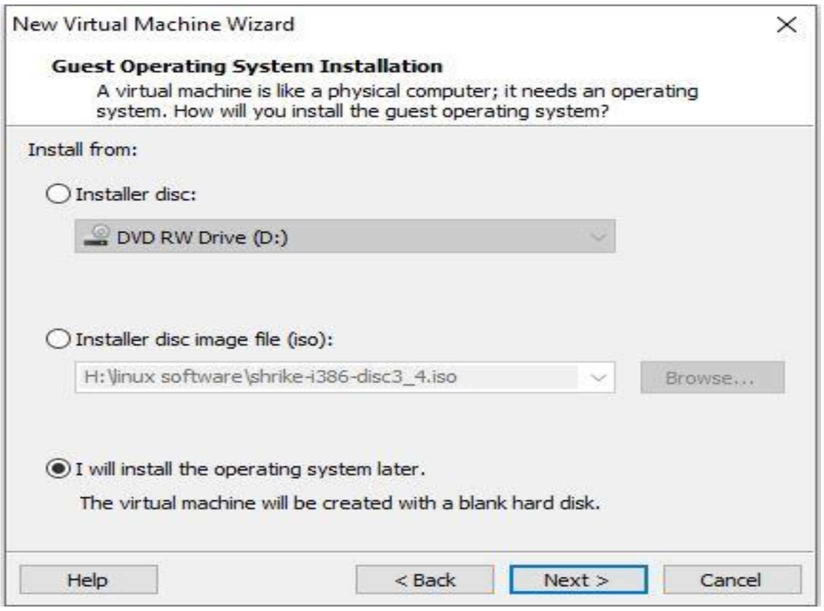
New Virtual Machine Wizard:

After the installation of VMWare in the system, the New Virtual Machine Wizard will open. It asks for the type of configuration you want to go with: typical or custom. The typical configuration is the recommended one and custom is the configuration with the advanced options. Choose the appropriate option and go to next step.

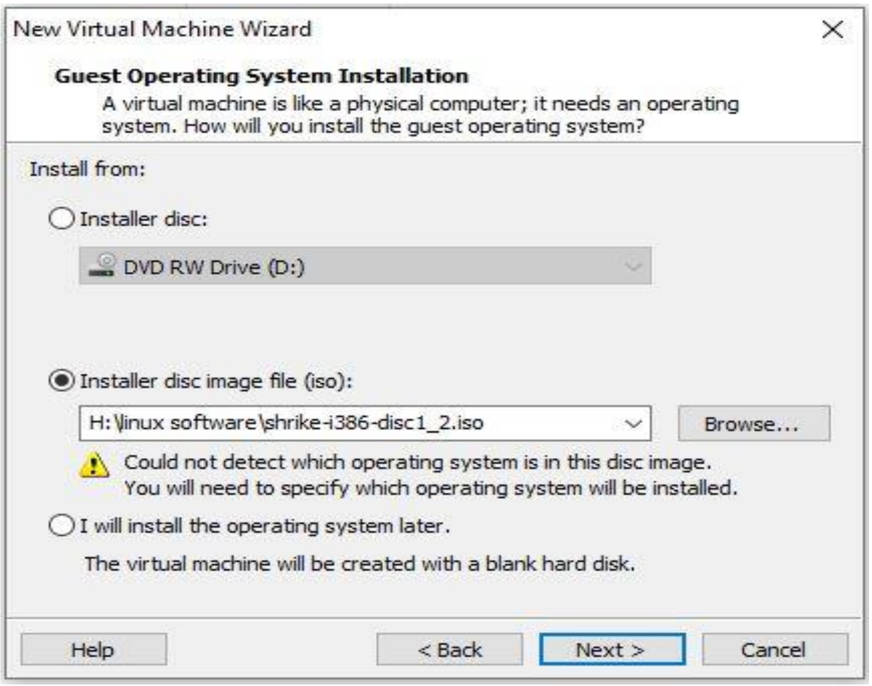
Installation of operating system

In this step, the guest operating system will be installed. Here the available options are

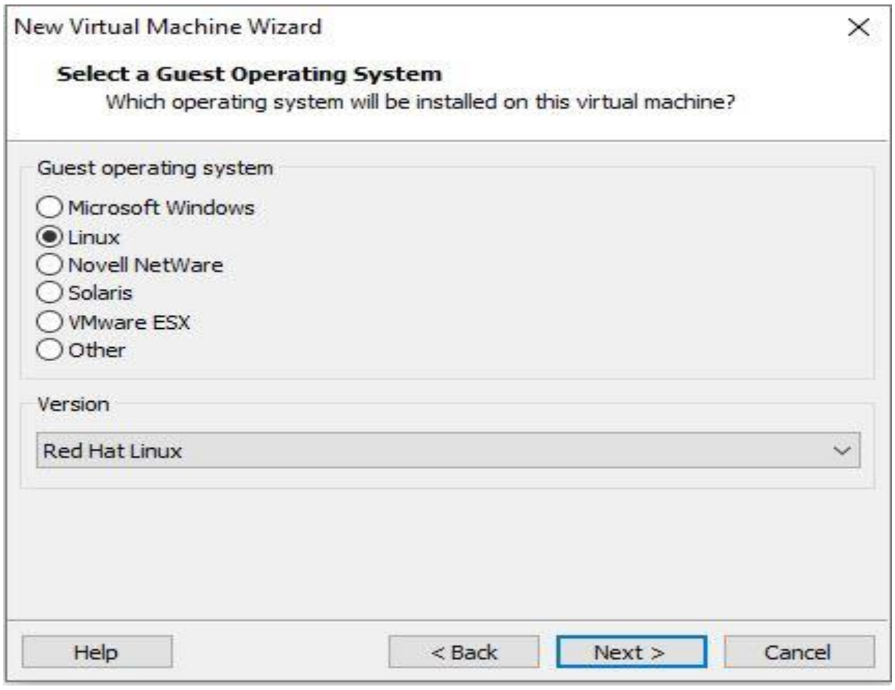
- Installer disc
- Installer disc image file (iso)
- Later installation.



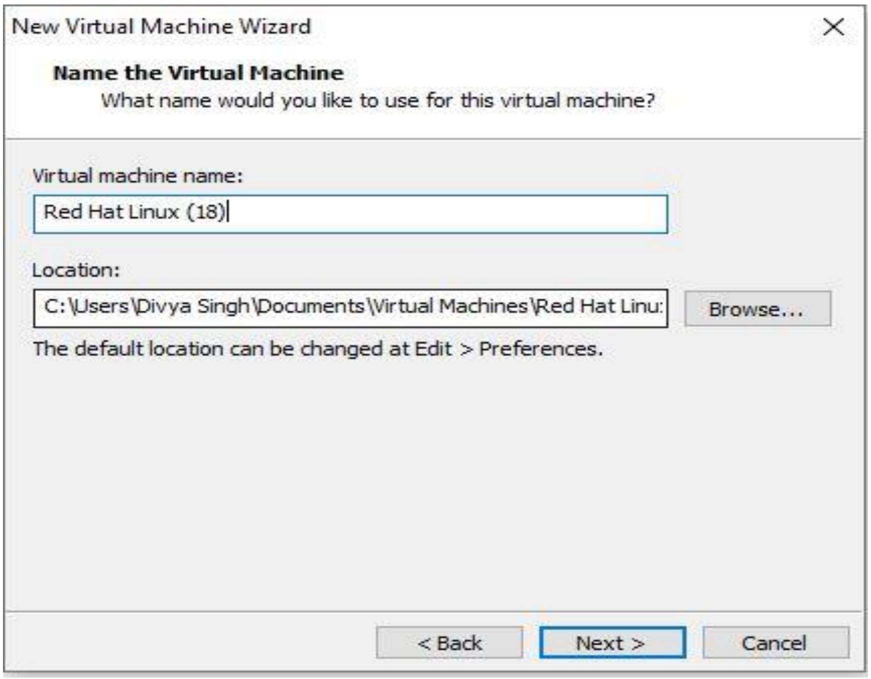
So out of these three options, choose the option depending upon the availability of the operating system. Here the installation using iso images is shown. The whole operating system is divided into three iso images.



After selecting the first iso image, choose which guest operating system you want to install. Here the Linux is chosen for installation on the virtual machine. The version of the operating system will also be selected here. Once it is done, click on Next to go further.

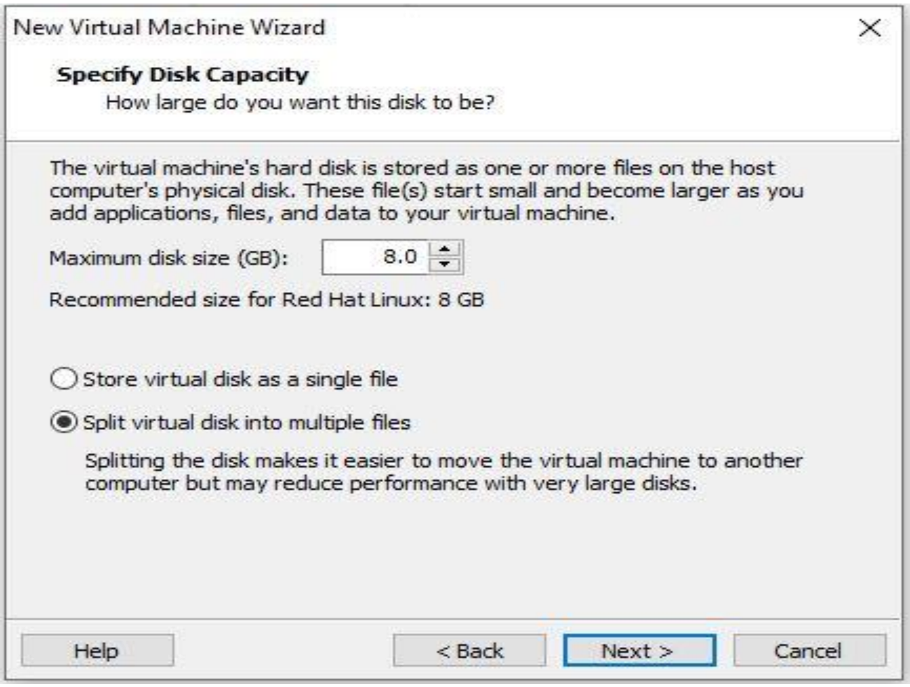


Write the name of the virtual machine and choose the location where you want to have all of its files. Once this is done, click on next to go further.



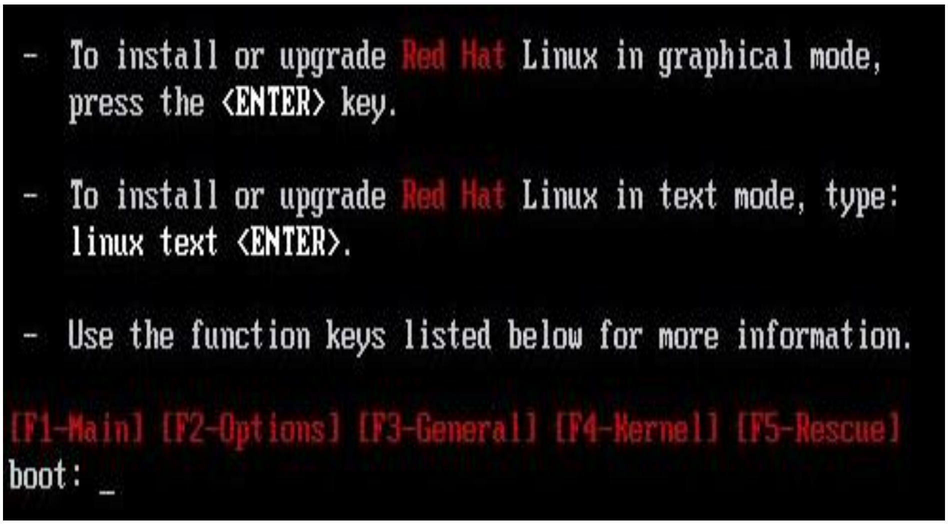
Specify disk capacity

When the name and location is specified, next task is to specify the disk capacity. By default, the recommended maximum disk size is 8 GB. We can increase or decrease it as per our requirements. Then it will ask whether you want to store virtual disk as a single file or into multiple files. Choose the appropriate option and go further.



After this, there are two options. One is to power on this virtual machine and other is to edit virtual machine settings. If you have done any mistake while setting the machine, then edit those settings. Otherwise turn on the power machine.

When you power on the virtual machine, the next screen will be:



When you press the ENTER key, the next screen will be:

Testing the CD Media

If you are using the CD media for the first time, then it is advised to test it before the installation process. If the same media is used for installation earlier, then it can be skipped.

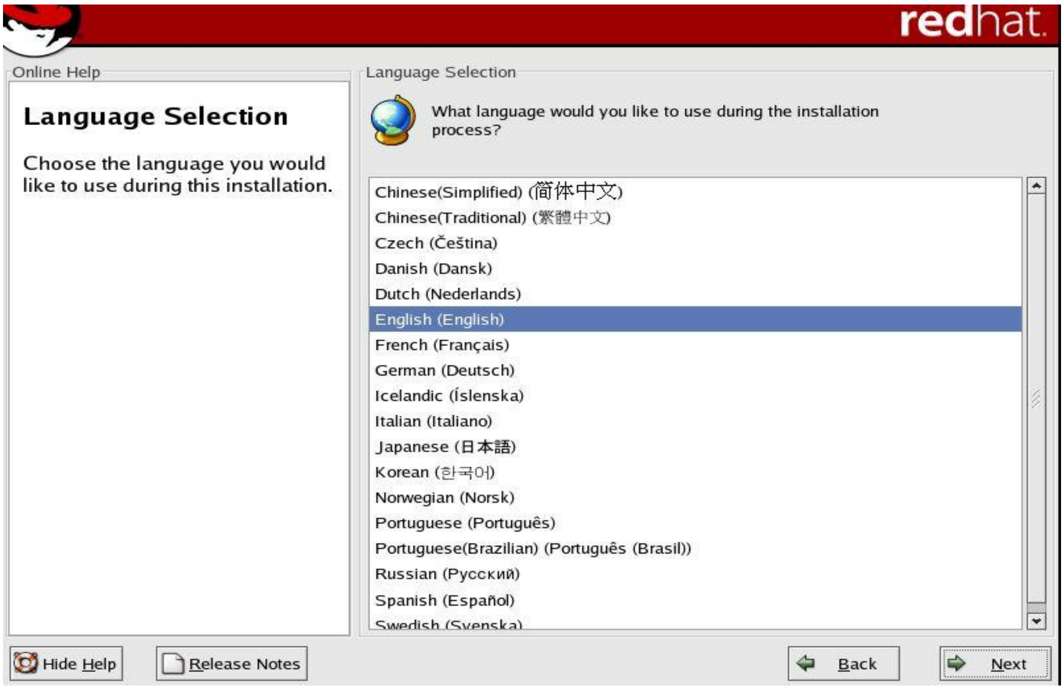
Running Anaconda

The Anaconda starts running. It is best to install Anaconda for the local user, which does not require administrator permissions and is the most robust type of installation. However, if you need to, you can install Anaconda system wide, which does require administrator permissions.

The process starts with the screen "Welcome to Red Hat Linux". Click on Next for further process.

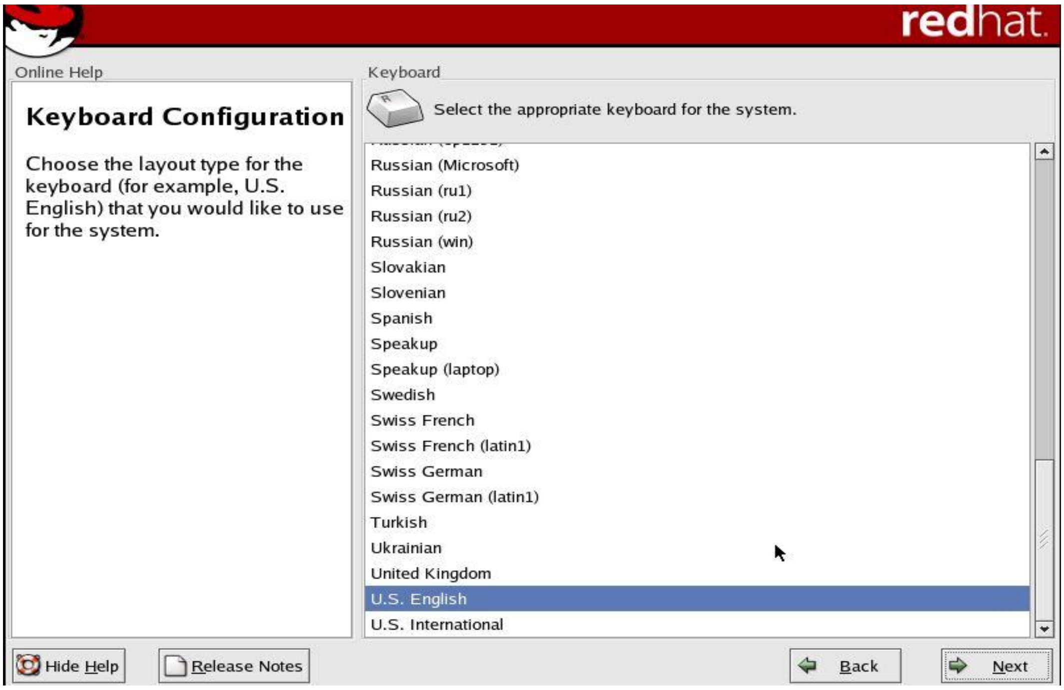
Language Selection

The next screen is for language selection. Choose the language that you want to use during installation process. After selection of language, click on next.



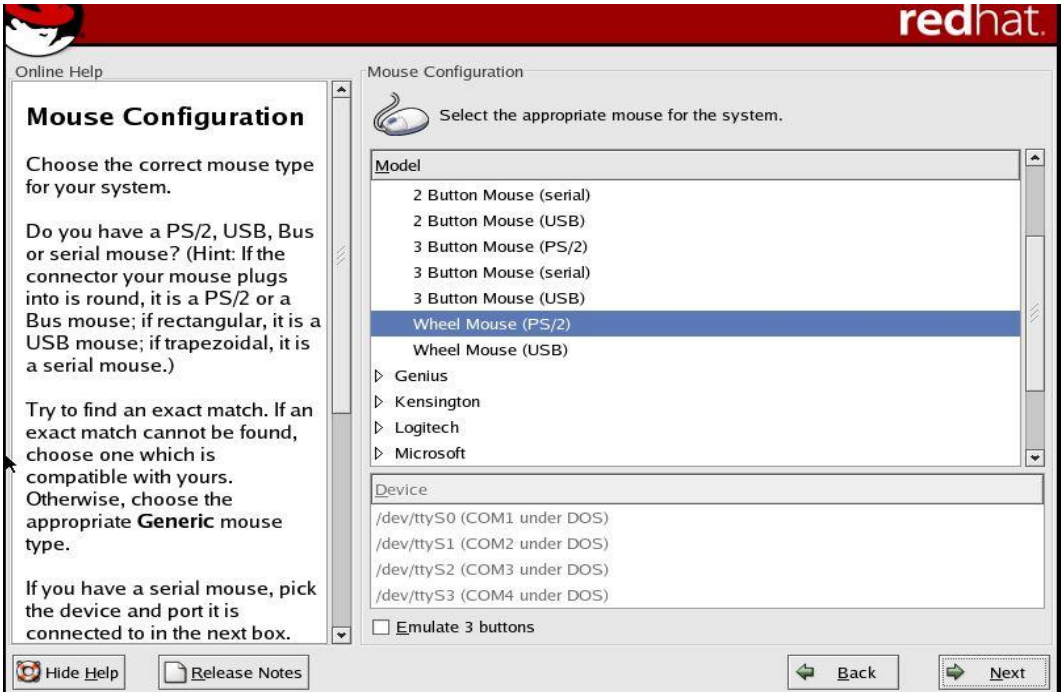
Keyboard Configuration

The screen is for choosing of layout type of keyboard that we want to use for the system. After chosen, click on next.



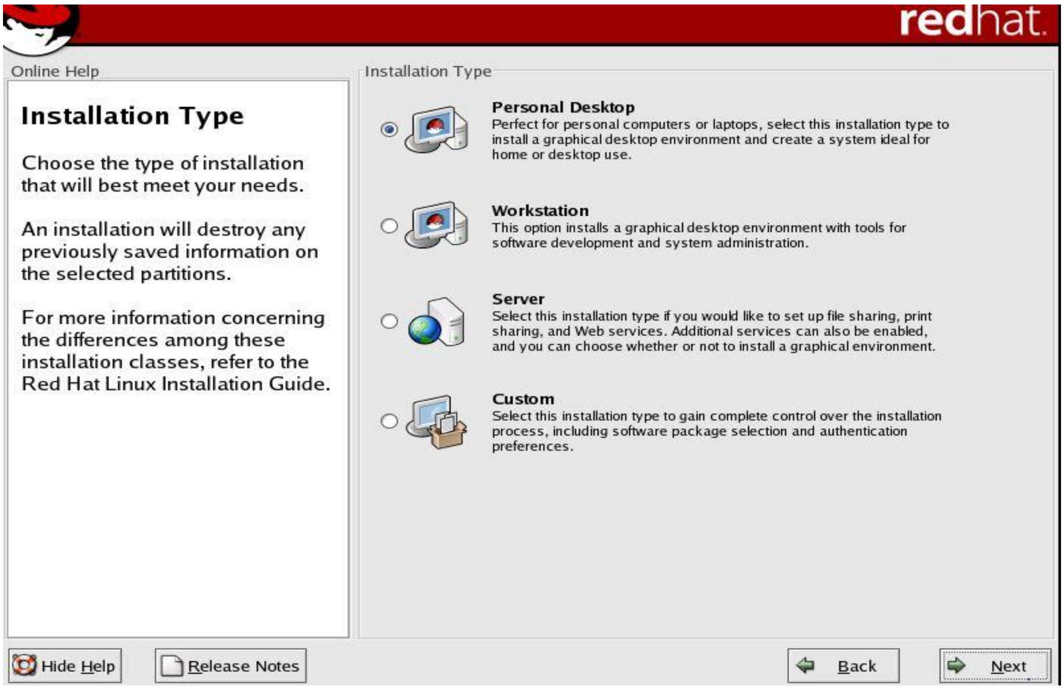
Mouse Configuration

The screen is for mouse configuration. By default, it will choose the appropriate option. So, it is advised to go with the recommended settings. Click on next.



Installation Type

Next is to choose the installation type which best meet your needs. The installation type available are: Personal Desktop, workstation, server and custom. Choose the appropriate option and click on next.



Disk Partitioning Setup

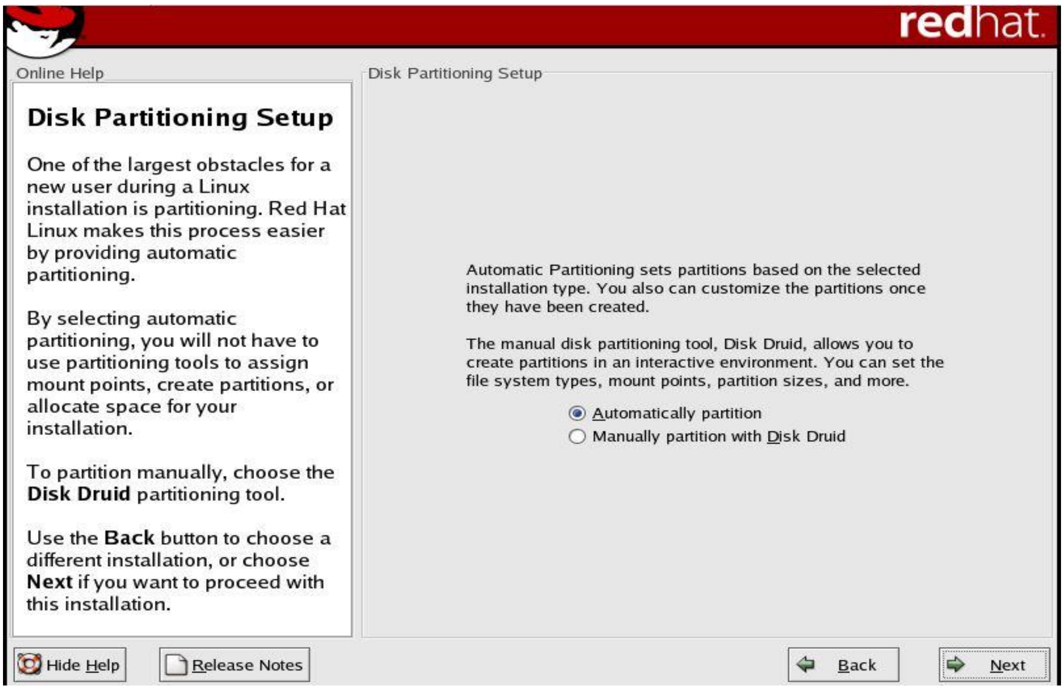
Partitioning is a means to divide a single hard drive into many logical drives. A partition is a contiguous set of blocks on a drive that are treated as an independent disk. A partition table is an index that relates sections of the hard drive to partitions.

The task is now to partition the drive. It can be done in two ways: automatically partition and manually partition with disk druid. The automatically partition is dependent upon the selected installation type. The manual partition uses the tool Disk Druid for partitioning. The ppartition fields are:

- **Device:** This field displays the partition's device name.
- **Start:** This field shows the sector on your hard drive where the partition begins.
- **End**: This field shows the sector on your hard drive where the partition ends.
- **Size**: This field shows the partition's size (in MB).
- **Type:** This field shows the partition's type (for example, ext2, ext3, or vfat).
- **Mount Point**: A mount point is the location within the directory hierarchy at which a volume exists; the volume is "mounted" at this location. This field indicates where the partition will be mounted.

The file system types are:

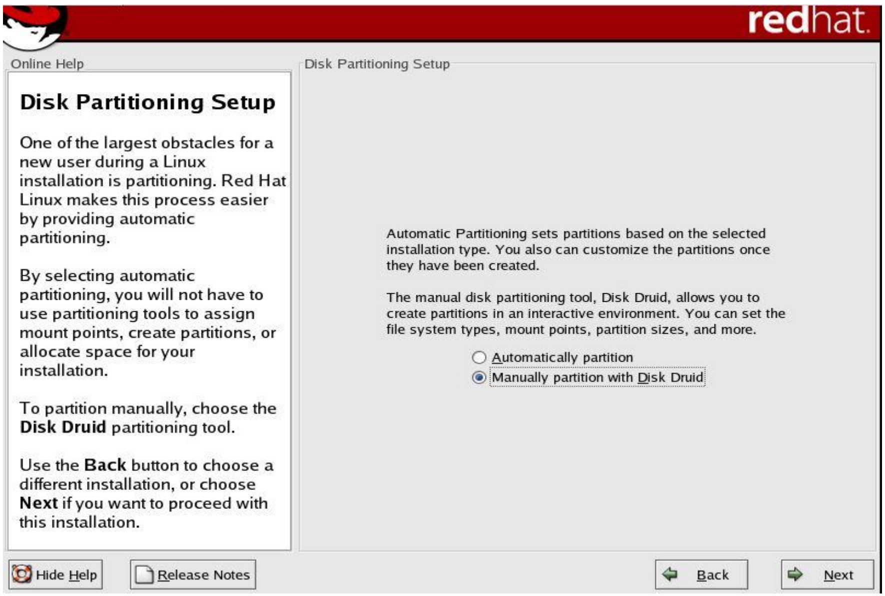
- **ext2 —** An ext2 filesystem supports standard Unix file types (regular files, directories, symbolic links, etc). It provides the ability to assign long file names, up to 255 characters. Versions prior to Red Hat Linux 7.2 used ext2 file systems by default.
- **ext3 —** The ext3 filesystem is based on the ext2 filesystem and has one main advantage — journaling. Using a journaling filesystem reduces time spent recovering a filesystem after a crash as there is no need to fsck the filesystem.
- **swap —** Swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing.
- **vfat —** The VFAT filesystem is a Linux filesystem that is compatible with Windows 95/NT long filenames on the FAT filesystem.

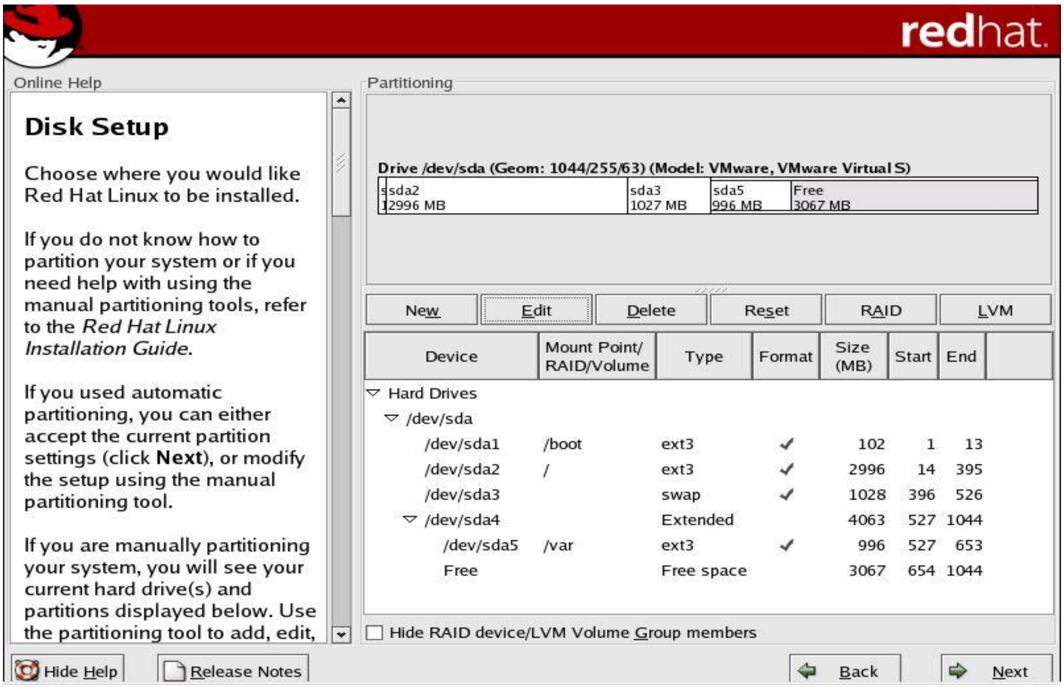


Recommended Partitioning Scheme

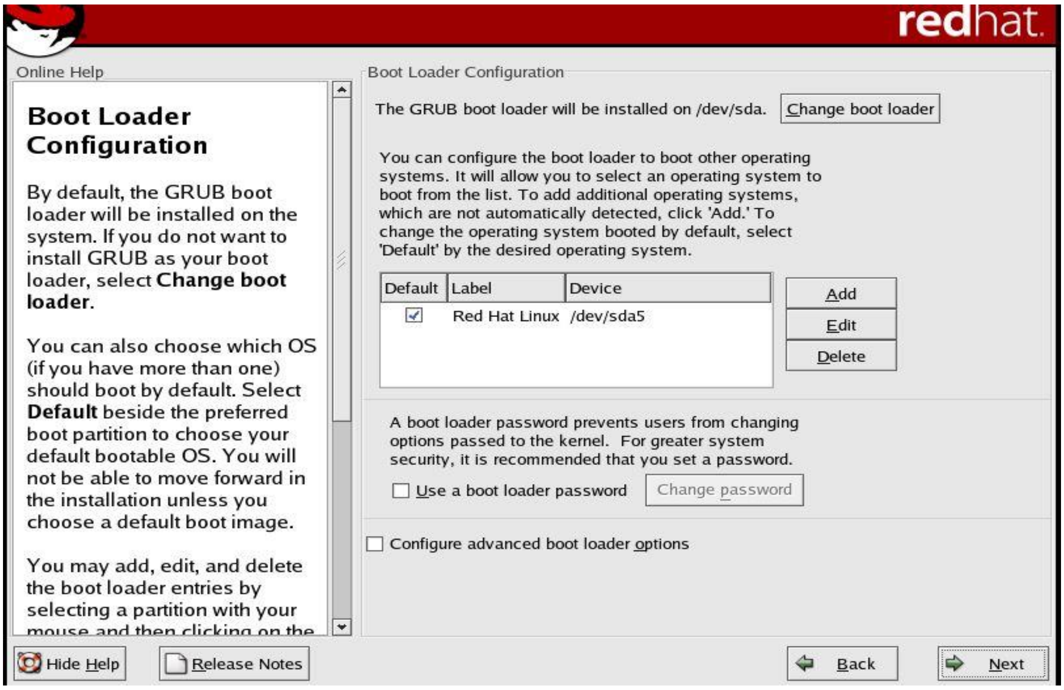
- Unless you have a reason for doing otherwise, it is recommended that you create the following partitions:
- /boot partition – contains kernel images and grub configuration and commands
- / partition
- /var partition
- Any other partition based on application (e.g /usr/local for squid)
- swap partition — swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing. The size of your swap partition should be equal to twice your computer's RAM.

Here we are going with manual partition in which disk druid tool helps.

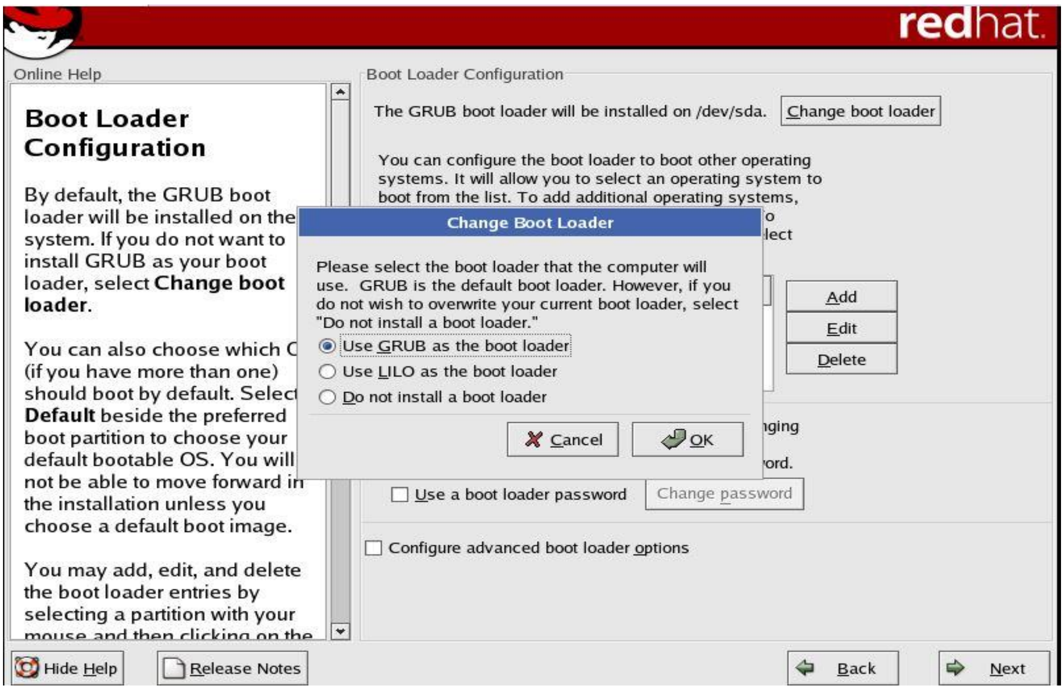


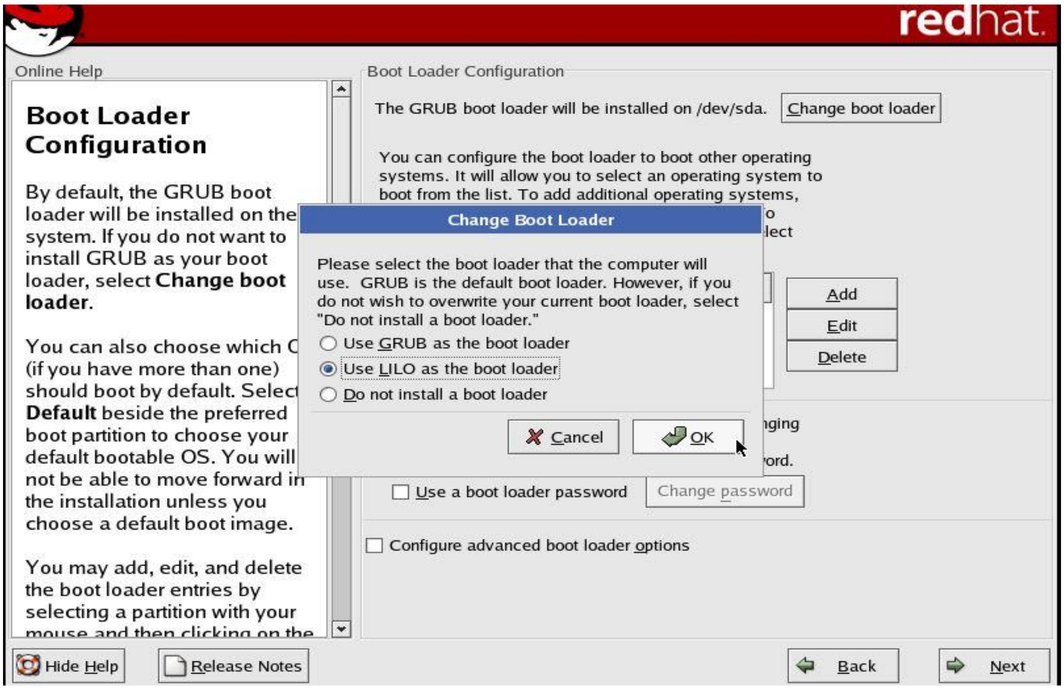


Boot Loader Configuration

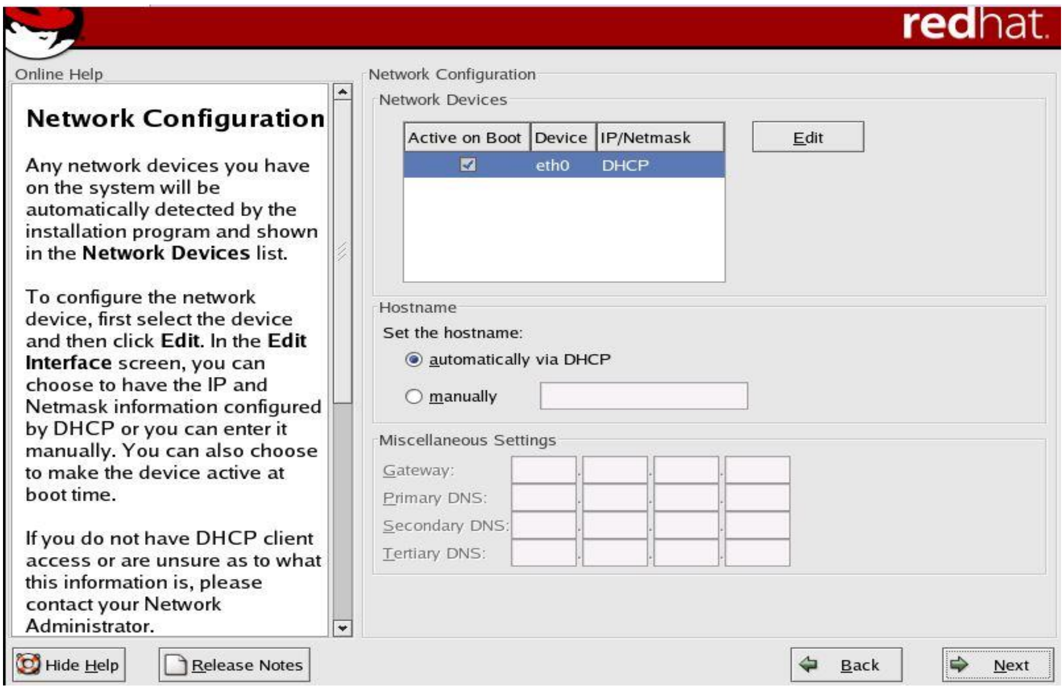


Change Boot Loader

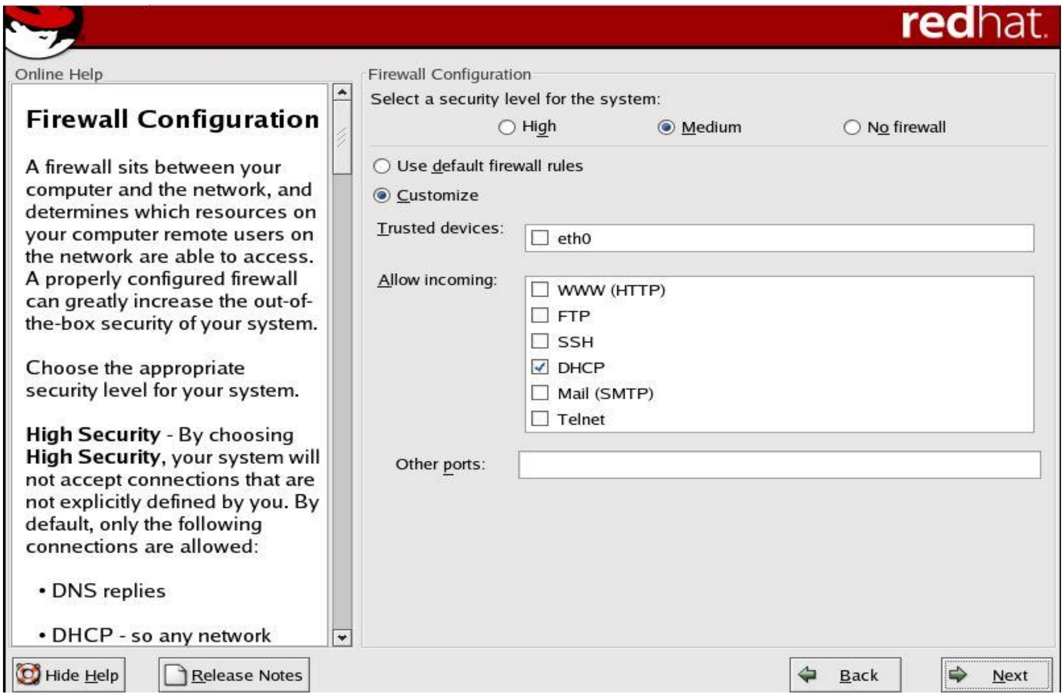




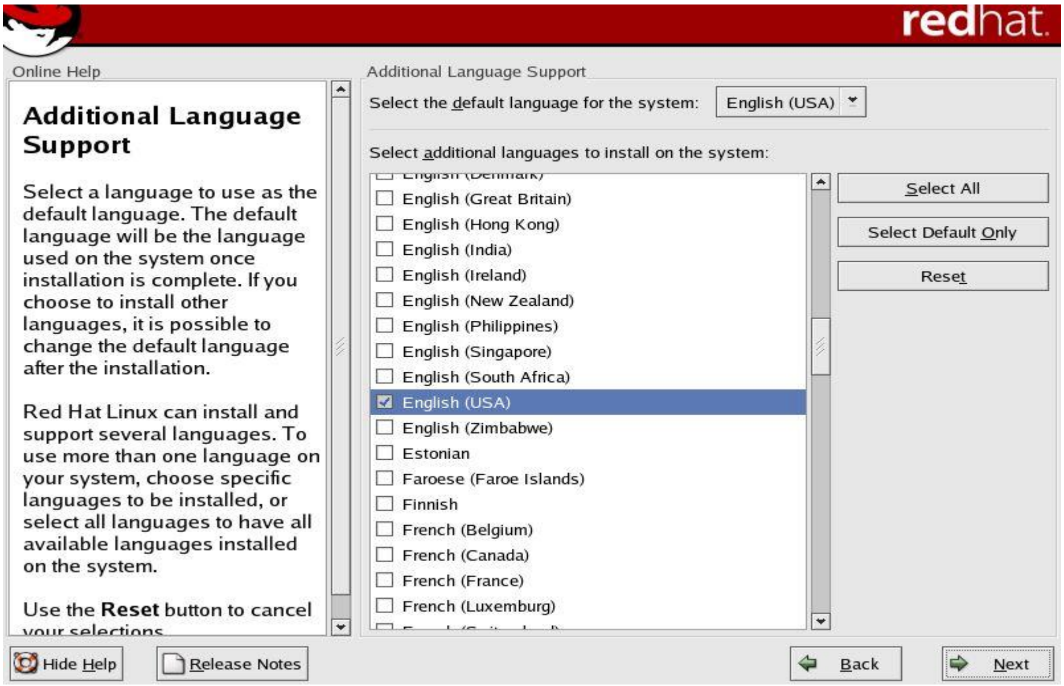
Network Configuration



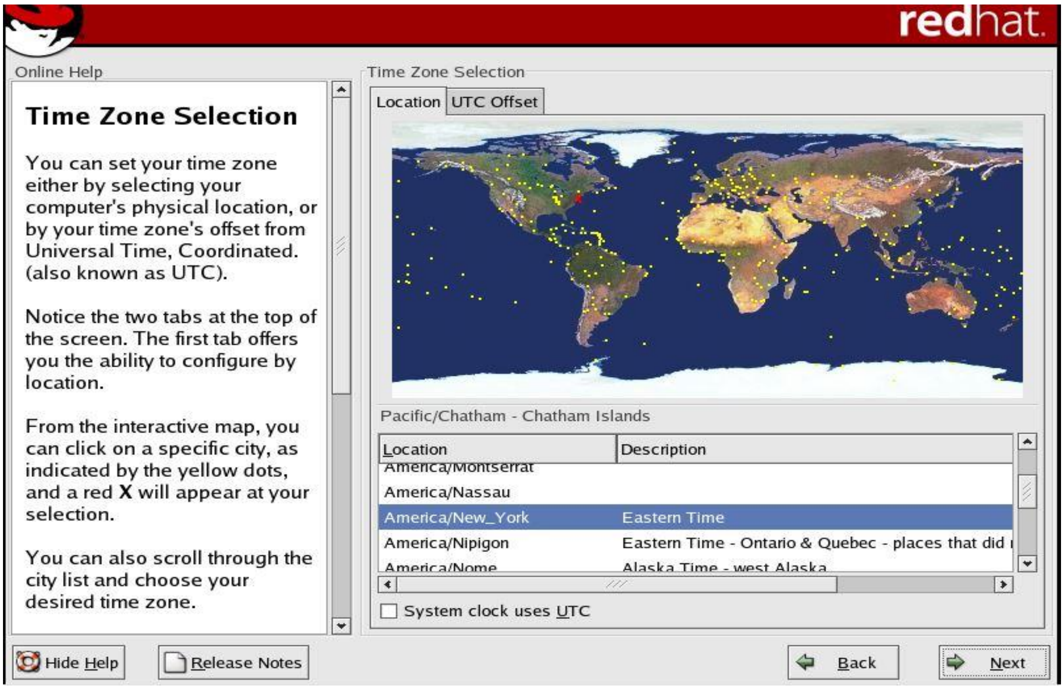
Firewall Configuration



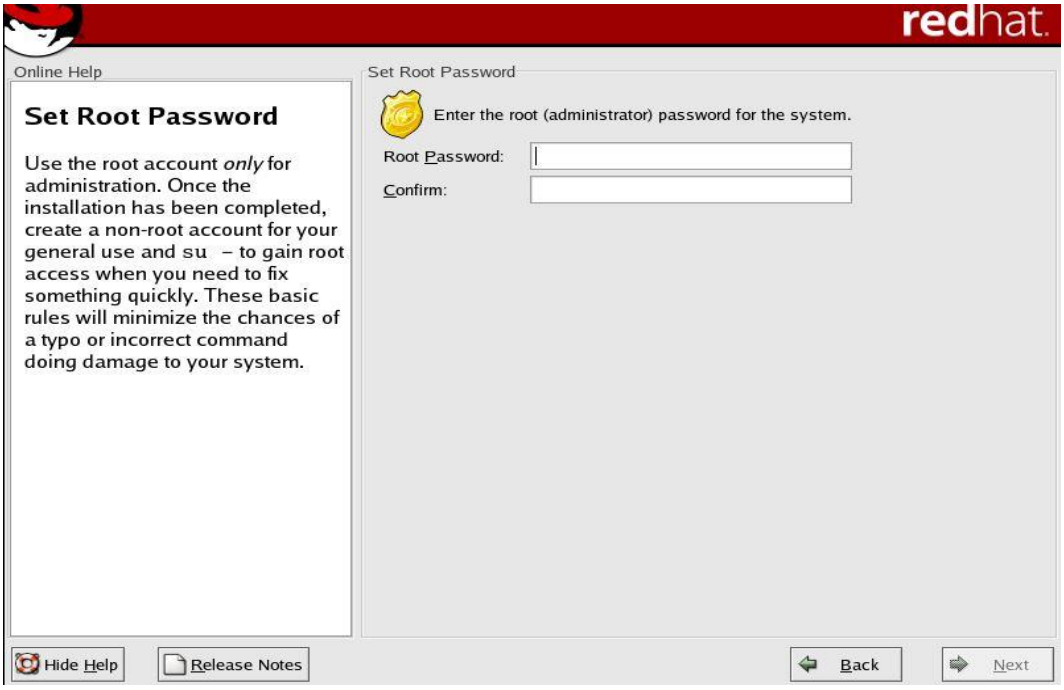
Additional Language Support



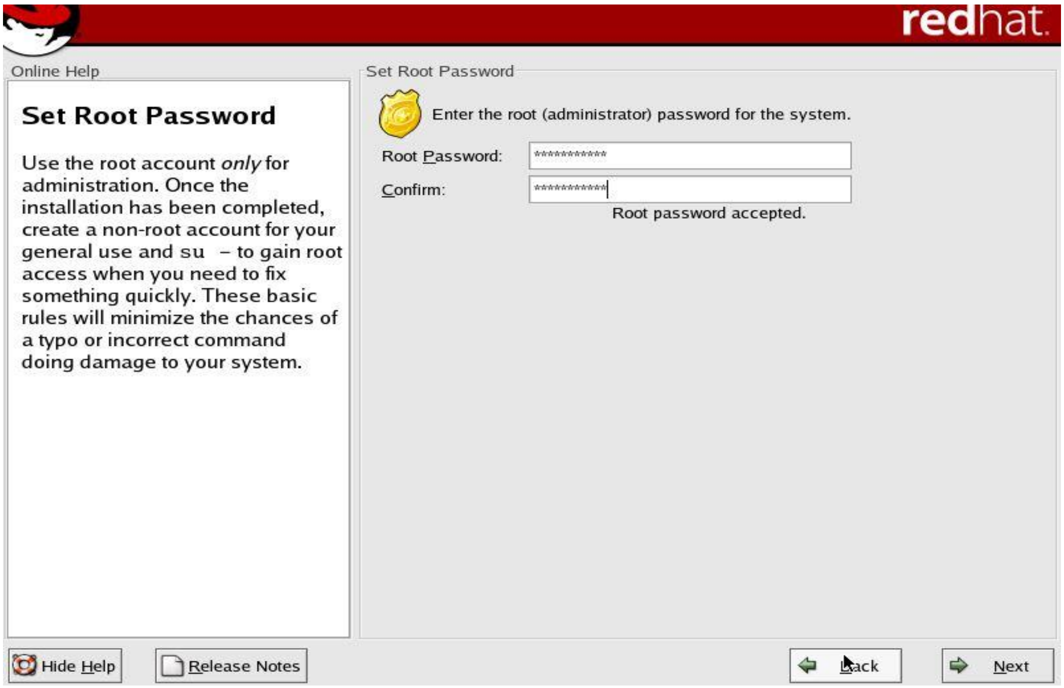
Time Zone Selection



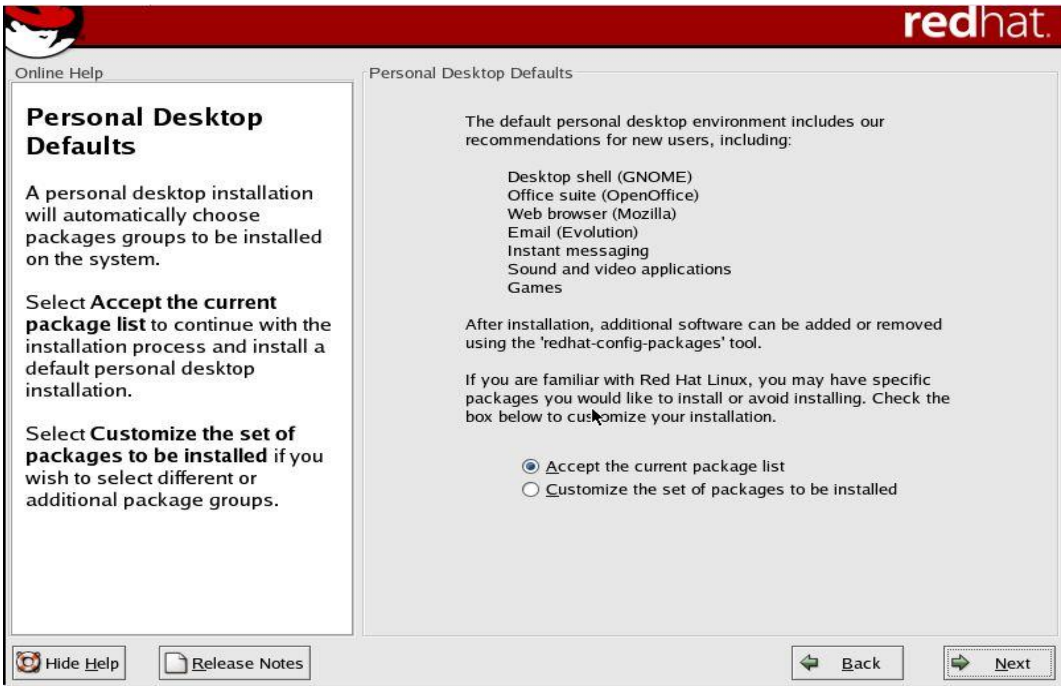
Set Root Password



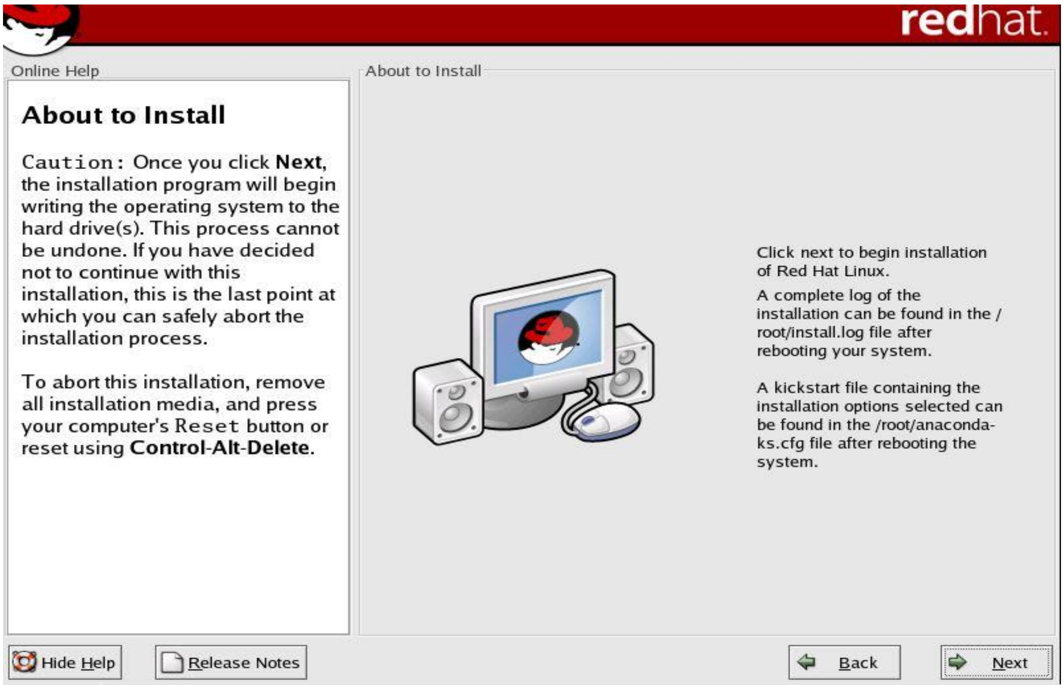
Provide Root Password



Personal Desktop Defaults (Packages)

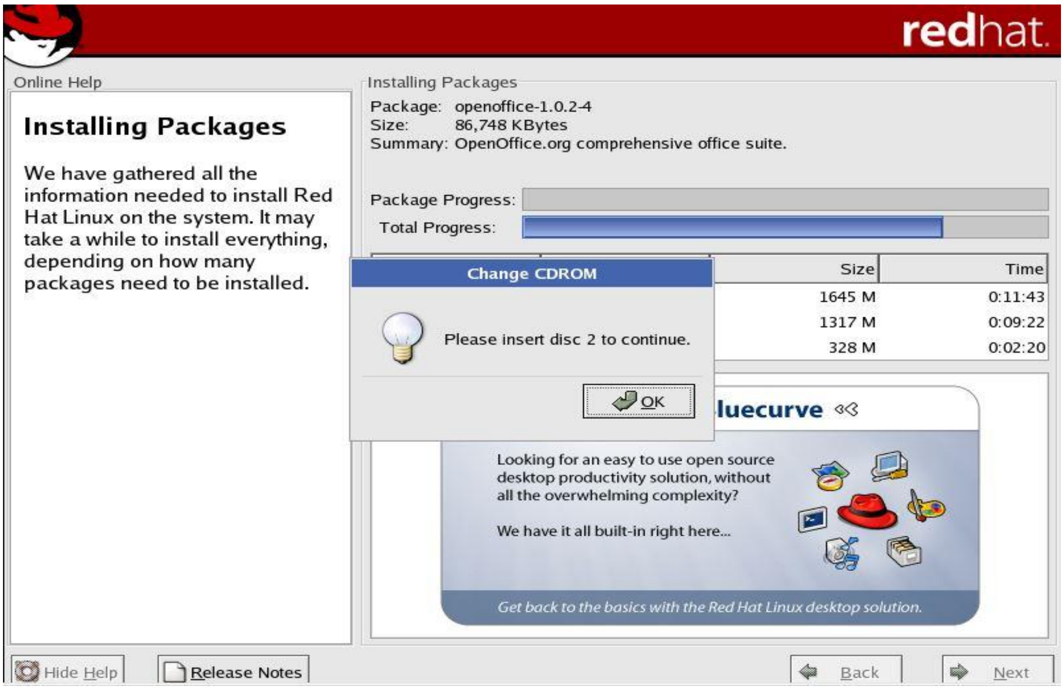


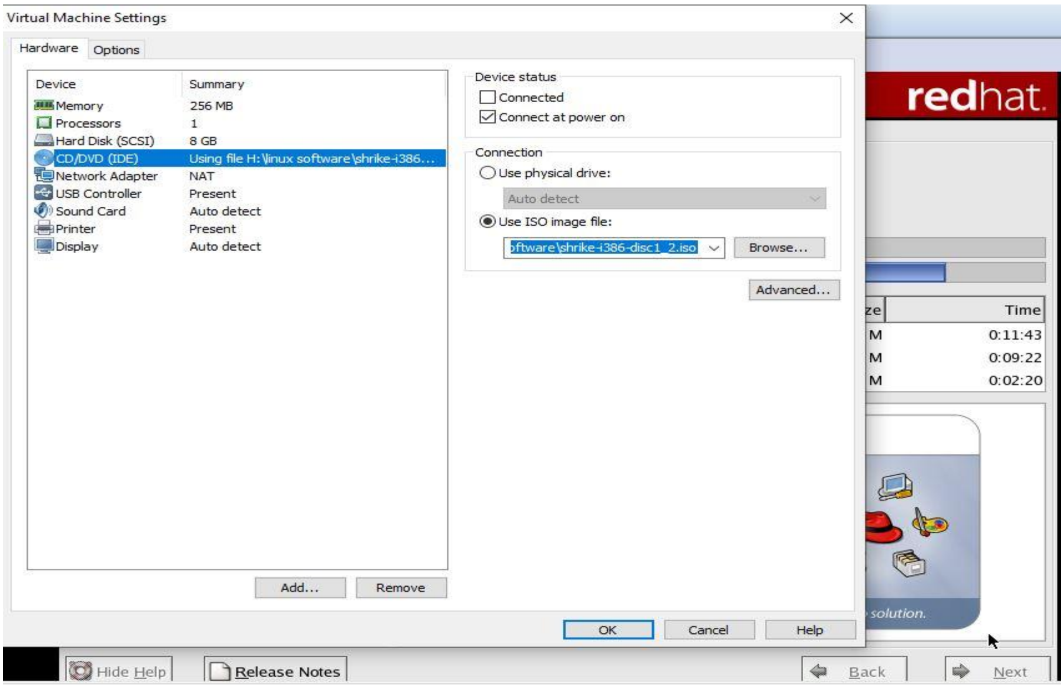
About to Install



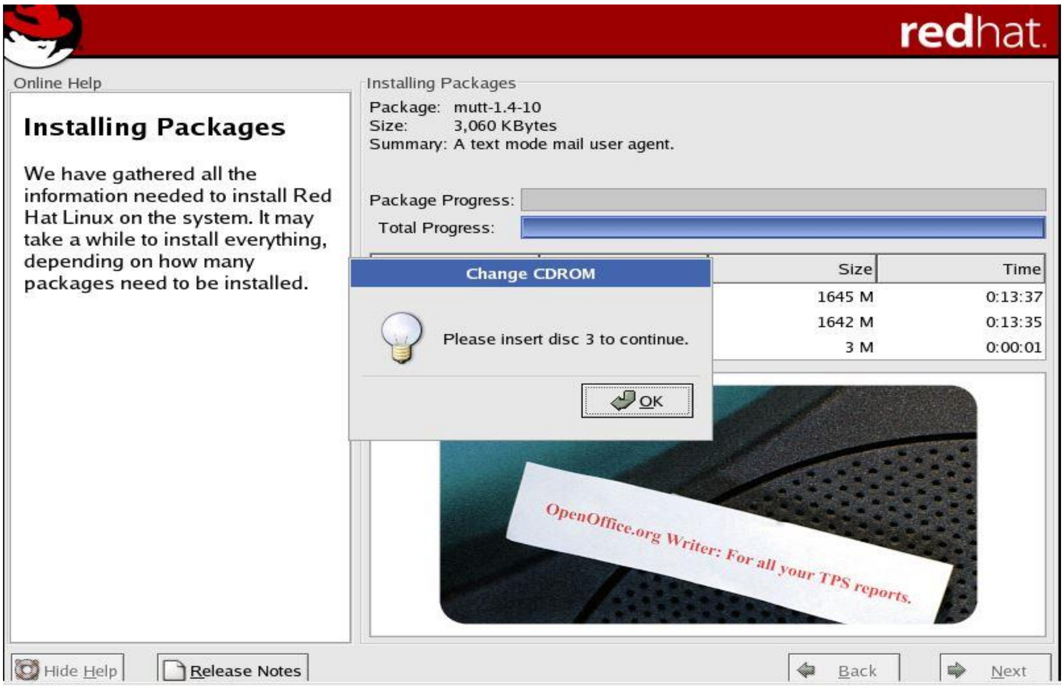
Installing Packages

Insert Disc 2

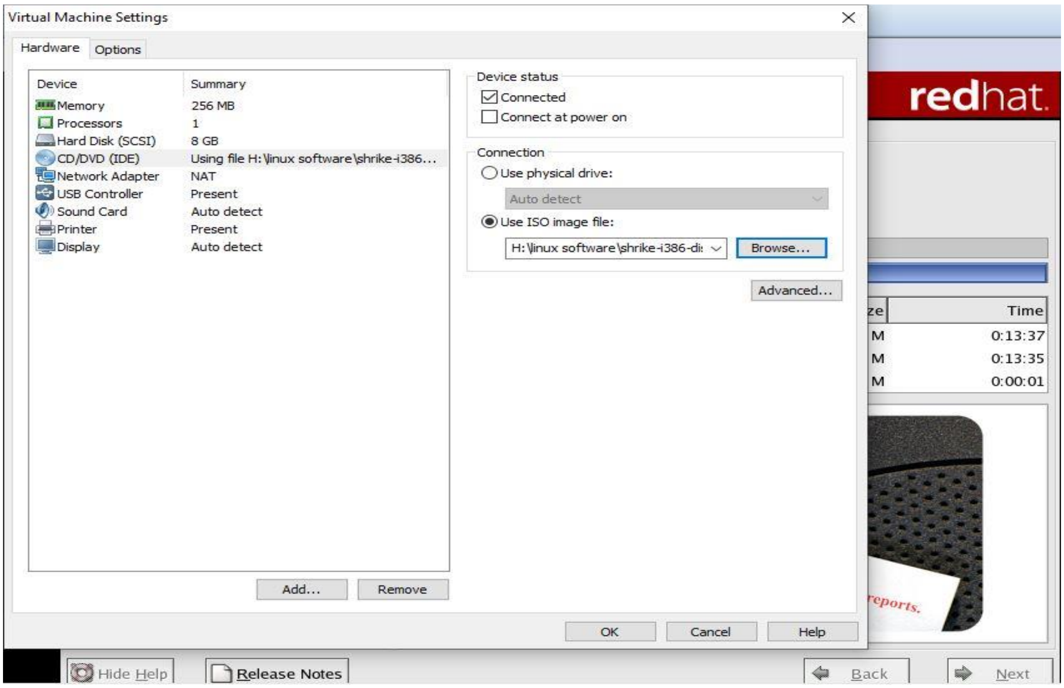




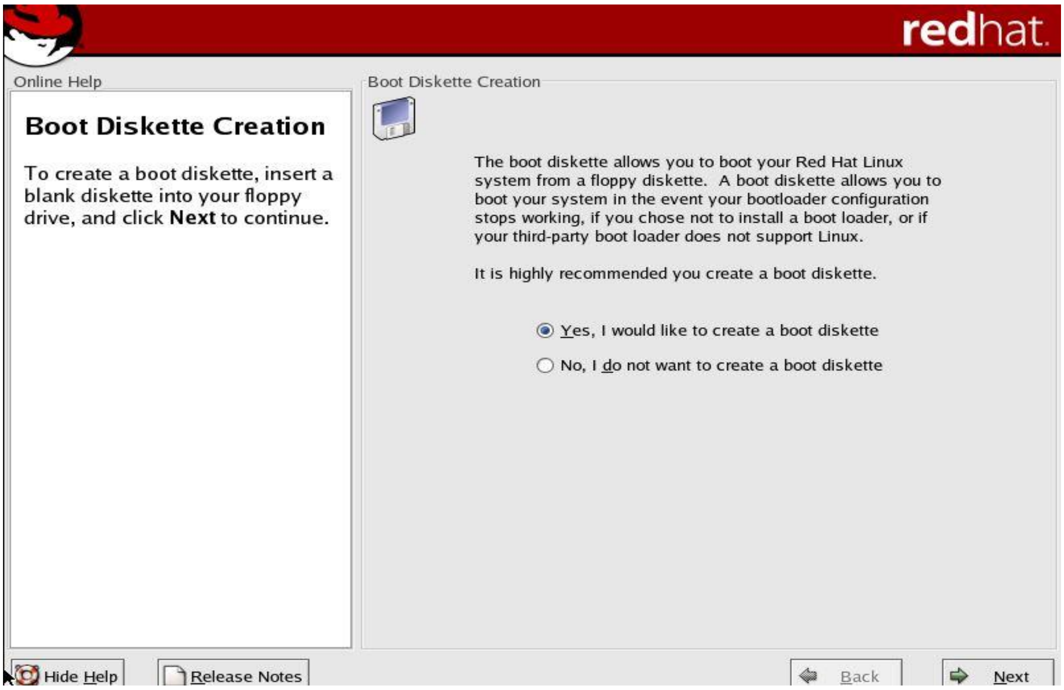
Insert Disc 3



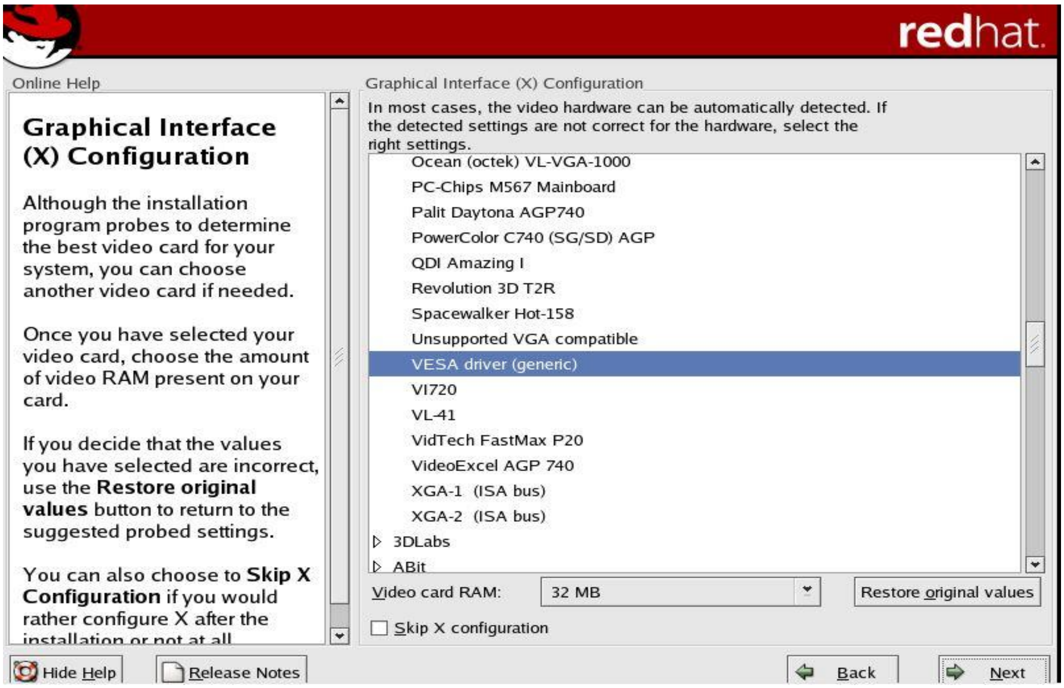
Proper Settings



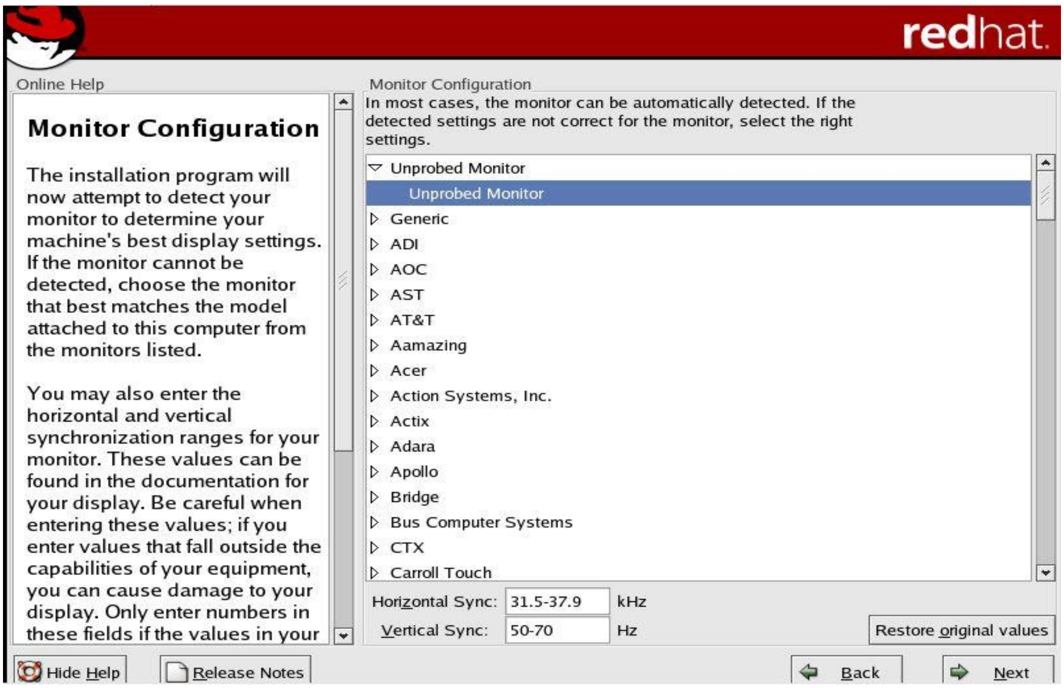
Boot Diskette Creation



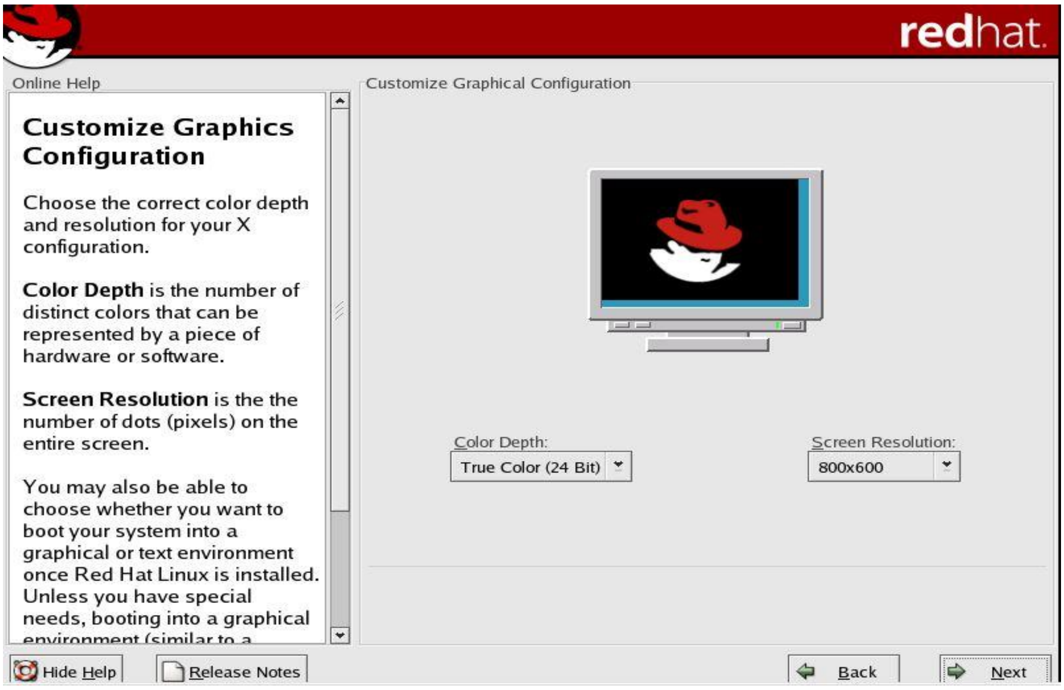
Graphical Interface (X) Configuration

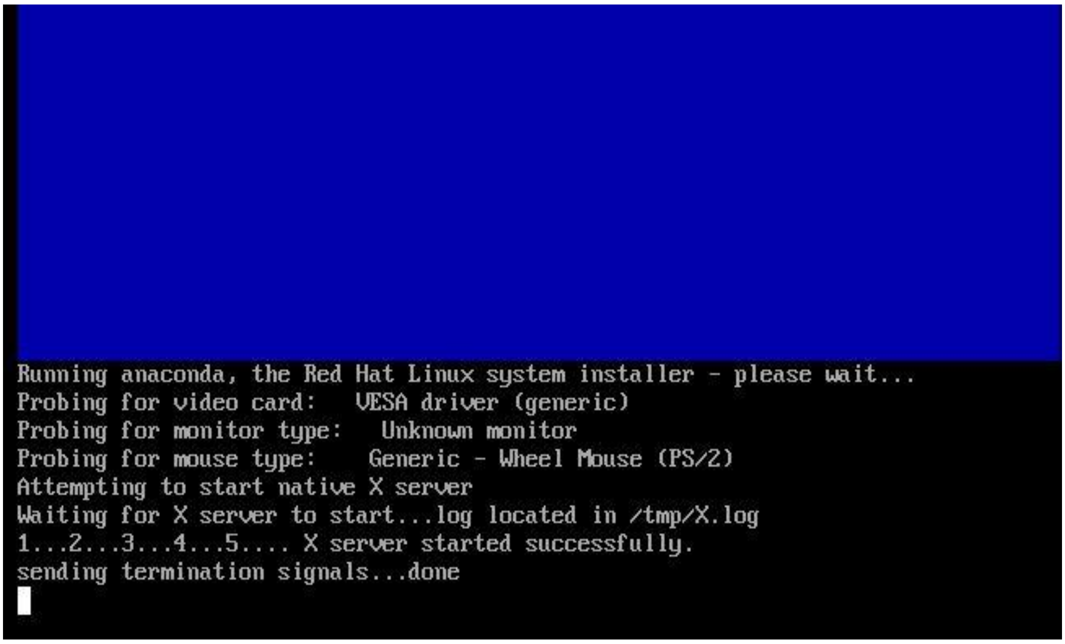


Monitor Configuration



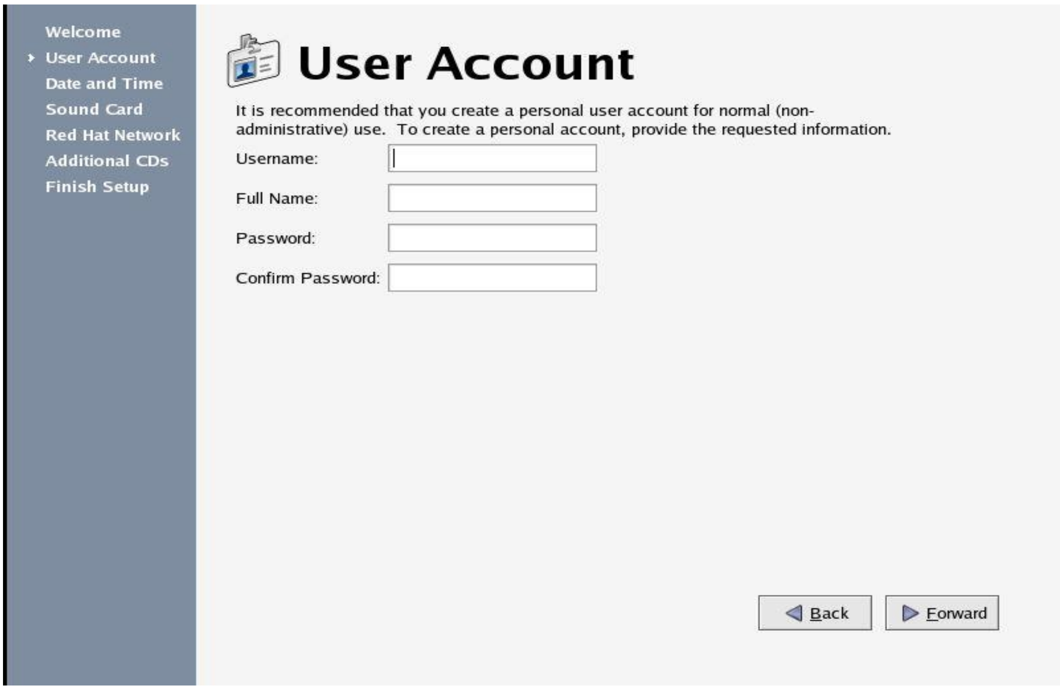
Customize Graphics Configuration





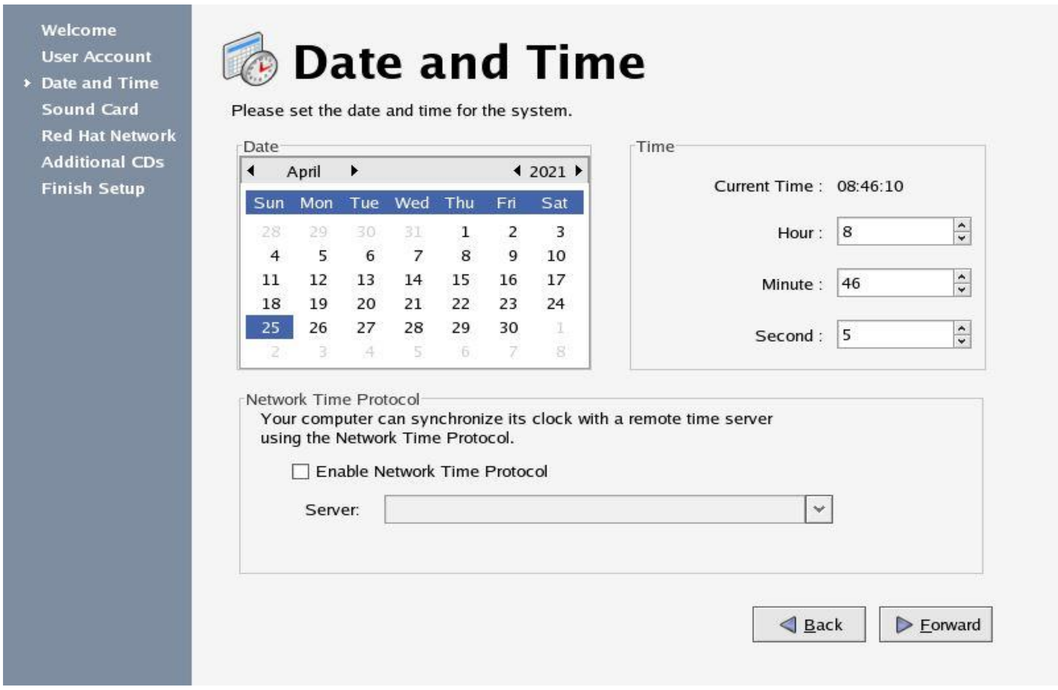
Welcome Screen

Creation of User Account



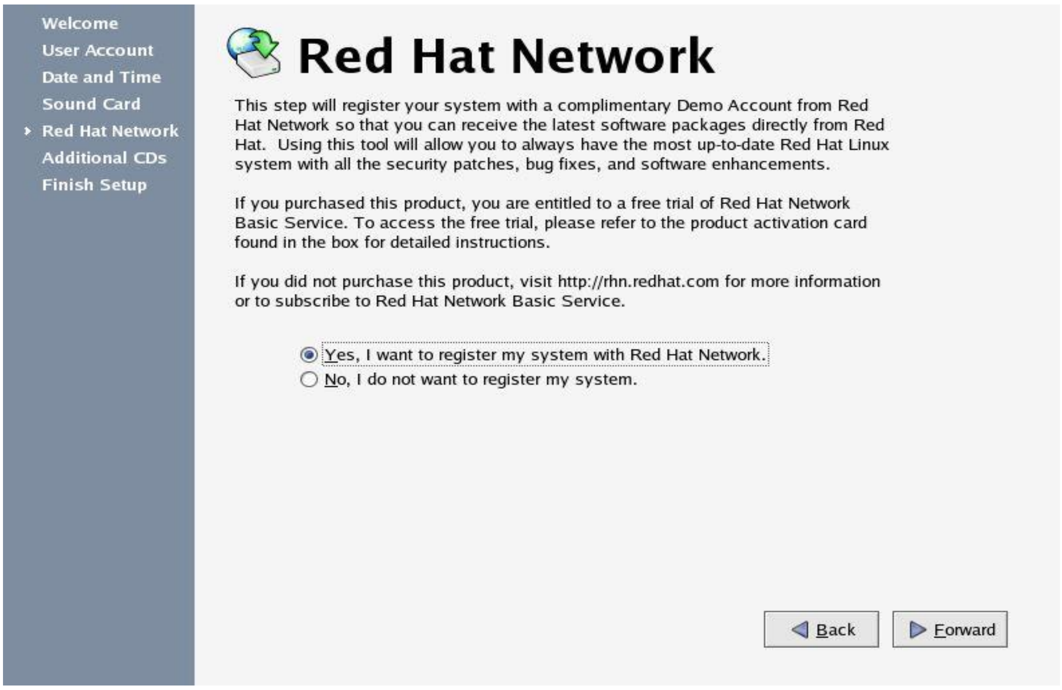
Enter Credentials

Enter Date and Time



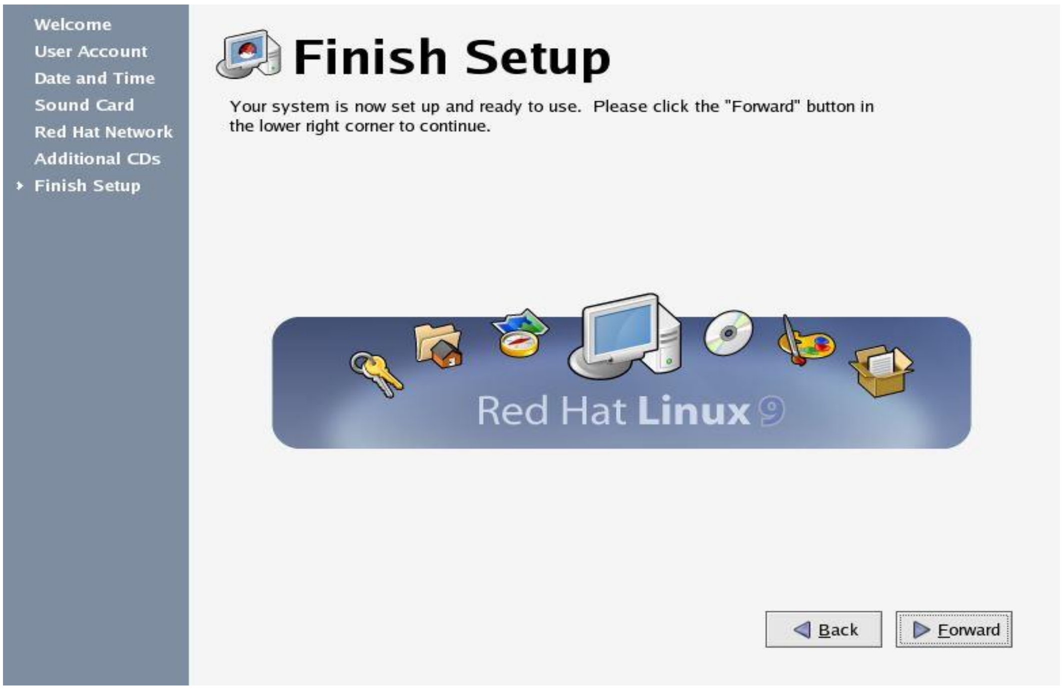
Sound Card

Registration with Red Hat Network



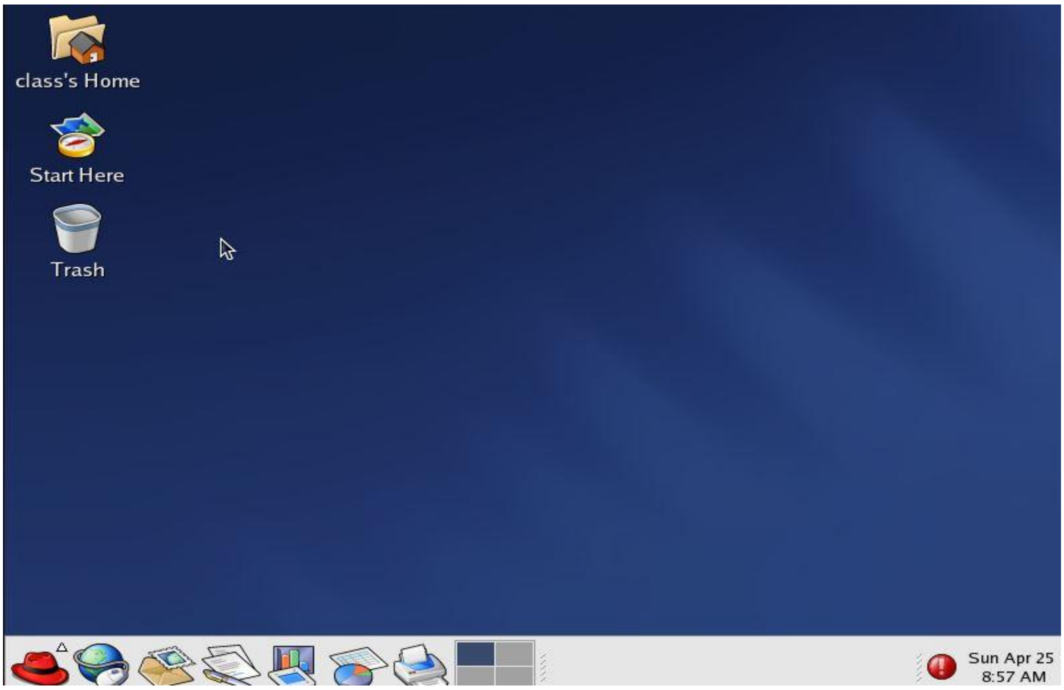
Installation of packages from additional CDs

Finish Setup



Logging in

Home Screen



2.4 Moving around the Desktop

- **GNOME**: The default desktop interface of Red Hat Linux 9. GNOME represents a presentation layer that provides a graphical user interface as well as the focused working environment, which enables you to access all your work from one place.
- **KDE**: KDE desktop is included in Rec Hat Linux 9 distribution but not installed by default.KDE is a desktop environment for an integrated set of cross-platform applications designed to run on Linux, FreeBSD, Microsoft Windows, Solaris and Mac OS, designed by the KDE Community.

2.5 Components of Desktop

- Panel: The panel stretches across the bottom of the desktop. By default, it contains the main menu icon and quick-launch icons for logging out, opening a terminal window, and other common applications and utilities. The panel is highly configurable. You can add and remove buttons that launch applications easily.
- Workspace: Workspaces refer to the grouping of windows on your desktop. You can create multiple workspaces, which act like virtual desktops. Workspaces are meant to reduce clutter and make the desktop easier to navigate.Workspaces can be used to organize your work. For example, you could have all your communication windows, such as e-mail and your chat program, on one workspace, and the work you are doing on a different workspace. Your music manager could be on a third workspace.

Summary:

- A boot sequence is the set of operations the computer performs when it is switched on that load an operating system.
- The primary function of BIOS is code program embedded on a chip that recognizes and controls various devices that make up the computer.
- GRUB and LILO are the most popular Linux boot loader.
- The first thing the kernel does is to execute init program.
- There are four types of installation available in Linux: personal desktop, workstation, server and custom.
- There are two ways the disk can be partitioned: manually and automatically.

Keywords:

- Booting: It is a bootstrapping process that starts operating systems when the user turns on a computer system.
- BIOS: It refers to the software code run by a computer when first powered on.
- Partitioning: It is a means to divide a single hard drive into many logical drives.A partition is a contiguous set of blocks on a drive that are treated as an independent disk.
- Swap: Swap partitions are used to support virtual memory. In other words, data is written to a swap partition when there is not enough RAM to store the data your system is processing.
- GNOME: The default desktop interface of Red Hat Linux is GNOME.
- Run-level: A run-level is a software configuration of the system which allows only a selected group of processes to exist

Self Assessment

1. Which of these executes kernel?
A. MBR
B. BIOS
C. GRUB
D. Init

2. The first thing a Kernel does is __________.
A. Execute GRUB
B. Execute LILO
C. Execute init program
D. None of the above

3. Which of these tasks are handled by Kernel?
A. System Call
B. Process Management
C. Device Management
D. All: System call, process and device management

4. While installation of Red Hat Linux in the system, it asks for __________.
A. Language Selection
B. Keyboard Configuration
C. Mouse Configuration
D. All: Language selection, keyboard and mouse configuration

5. In partition field, i.e., SIZE, the measurement unit is _____
A. TB
B. MB
C. GB
D. KB

6. Which of these defines the runlevels?
A. 0-6
B. 1-7
C. 2-8
D. 3-9

7. Which of these partitions is used to support virtual memory?
A. /
B. /var
C. /boot
D. swap

8. Which of these is a Red Hat Linux installer?
A. Anaconda
B. GRUB
C. LILO
D. Emulator

9. Which of these programs allows us to partition the disk?
A. Anaconda
B. GRUB
C. Disk Druid
D. Disk Help

10. MBR is executed by ___________ .
A. BIOS
B. GRUB
C. Kernel
D. Init

11. In which mode, it is possible to install and upgrade Red Hat Linux?
A. Graphical
B. Text
C. Both graphical and text
D. None of the above

12. What does BIOS mean?
A. Basic Input/ Output Service
B. Basic Input/ Output System
C. Buffer Input/ Output System
D. Buffer Input/ Output Service

13. MBR is executed by ___________ .
A. BIOS
B. GRUB
C. Kernel
D. Init

14. In which mode, it is possible to install and upgrade Red Hat Linux?
A. Graphical
B. Text
C. Both graphical and text
D. None of the above

15. What type can be installed in Red Hat Linux?

A. Personal Desktop
B. Server
C. Workstation
D. All: personal desktop, server and workstation

Answers for Self Assessment

1. C	2. C	3. D	4. D	5. B
6. A	7. D	8. A	9. C	10. A
11. C	12. B	13. A	14. C	15. D

Review Questions:

1. What is booting? Explain the booting sequence in detail.
2. What is a kernel? Explain the tasks of a kernel in detail.
3. What is a partition? Write the partition fields. What is the recommended partition scheme?
4. What is a file system? Explain its types in detail.
5. What is a run-level? Explain about the run-level of in it.

Further Readings

Mark G. Sobell, A Practical Guide to Fedora andRed Hat Enterprise Linux, Fifth Edition, Pearson Education, Inc.

Web Links

https://www.educba.com/install-linux/

https://phoenixnap.com/kb/linux-create-partition